MSFMamba: Multiscale Feature Fusion State Space Model for Multisource Remote Sensing Image Classification

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***Abstract*— In the field of multisource remote sensing image classification, remarkable progress has been made by using the convolutional neural network (CNN) and Transformer. While CNNs are constrained by their local receptive fields, Transformers mitigate this issue with their global attention mechanism. However, Transformers come with the tradeoff of higher computational complexity. Recently, Mamba-based methods built upon the state space model (SSM) have shown great potential for long-range dependence modeling with linear complexity, but they have rarely been explored for multisource remote sensing image classification tasks. To address this issue, we propose the Multi-Scale Feature Fusion Mamba (MSFMamba) network, a novel framework designed for the joint classification of hyperspectral image (HSI) and light detection and ranging (LiDAR)/synthetic aperture radar (SAR) data. The MSFMamba network is composed of three key components: the Multi-Scale Spatial Mamba (MSpa-Mamba) block, the Spectral Mamba (Spe-Mamba) block, and the fusion Mamba (Fus-Mamba) block. The MSpa-Mamba block employs a multiscale strategy to reduce computational cost and alleviate feature redundancy in multiple scanning routes, ensuring efficient spatial feature modeling. The Spe-Mamba block focuses on spectral feature extraction, addressing the unique challenges of HSI data representation. Finally, the Fus-Mamba block bridges the heterogeneous gap between HSI and LiDAR/SAR data by extending the original Mamba architecture to accommodate dual inputs, enhancing cross-modal feature interactions and enabling seamless data fusion. Together, these components enable MSFMamba to effectively tackle the challenges of multisource data classification, delivering improved performance with optimized computational efficiency. Comprehensive experiments on four real-world multisource remote sensing datasets (Berlin, Augsburg, Houston2018, and Houston2013) demonstrate the superiority of MSFMamba outperforms several state-of-the-art methods and achieves overall accuracies of 76.92%, 91.38%, 92.38%, and 92.86%, respectively. The source codes of MSFMamba will be publicly available at https://github.com/oucailab/MSFMamba.**

***Index Terms*— Hyperspectral image (HSI), light detection and ranging (LiDAR), multisource data classification, state space model (SSM), synthetic aperture radar (SAR), Transformer.**

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I. Introduction

With the rapid development of remote sensing technology and sensor platforms, multisource remote sensing data from various platforms such as satellites, aircraft, and drones have become increasingly abundant [1], [2], [3]. These data have been widely applied in land use and land cover classification [4], geological resource exploration [5], urban planning [6], disaster warning [7], and agricultural monitoring [8], [9]. Among these applications, land use and land cover classification is an important and fundamental task [10], [11].

Among these multisource data captured by various sensors, hyperspectral image (HSI), light detection and ranging (LiDAR) data, and synthetic aperture radar (SAR) data are of great significance [12]. HSI can identify subtle differences among ground objects through its rich spectral information [13], [14], [15]. HSI classification has been widely studied since the rich spectral information contained in HSI presents great opportunities and challenges. On the other hand, LiDAR data precisely measure the 3-D structure of terrain and objects with high spatial resolution and accuracy [16], [17]. SAR sensors perform imaging in all-weather conditions and at all times, penetrating clouds and vegetation [18], [19], [20]. The fusion of HSI and LiDAR/SAR data can leverage the complementary advantages of multiple data sources and lead to improved ground object classification performance [21], [22], [23], [24]. By combining the spectral information of HSI with the 3-D structural information of LiDAR or the all-weather imaging capability of SAR, the limitations of a single data source can be alleviated, and more reliable classification results can be achieved [25], [26]. Therefore, in this article, we mainly focus on HSI and LiDAR/SAR data joint classification.

Because of diverse distributions and feature representations in multisource images, the critical challenge for HSI and LiDAR/SAR data classification lies in how to bridge

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the heterogeneity gap. Recently, tremendous efforts have been made for multisource data classification [27], [28], [29], [30].The most common framework that demonstrated promising classification results employs CNN-based feature encoder in an end-to-end manner. Cross-guided attention [31], cross-channel correlation [32], federated learning [33], and cross-scale mixing attention [34] are employed for multisource data classification. CNN-based methods design complex attention mechanisms to capture useful features, but, due to the inherent limitations of convolutional units, their receptive field range is limited, making it difficult to capture long-range dependencies. In addition, Transformer-based models have demonstrated outstanding long-range feature modeling capabilities in multisource remote sensing data classification [35]. Spatial–spectral attention [36], hierarchical attention [37], [38], and scale-adaptive attention [39] are used for HSI and LiDAR/SAR data classification. Although the Transformer has a global receptive field, its attention mechanism has high computational complexity and is less efficient when handling long-range dependencies.

Recently, an improved SSM [40] with a selective scanning mechanism, Mamba [41], has emerged as an effective alternative for computer vision and remote sensing image interpretation. As shown in Fig. 1, the Mamba-based method [42], [43] performs better than CNN and Transformer-based methods in effective receptive field modeling. By expanding the receptive field, the model is able to capture broader spatial dependencies, which is crucial for recognizing contextual objects and larger patterns in remote sensing data. In contrast to Transformer-based models, the state space sequence model reformulates the attention mechanism to scale linearly with sequence length, significantly reducing computational costs, especially for long sequences [44].

It is a nontrivial and challenging task to introduce the SSM into multisource data classification due to the following reasons.

1) *Feature redundancy in multiple scanning routes:* To address the challenge of image data nondirectionality, existing SSM-based models generally use a multiscan strategy (e.g., forward, backward, horizontal, and vertical scan) to ensure that every part of the image can establish connections with other parts. However, the multiscan strategy significantly increases feature redundancy, as similar patterns are repeatedly extracted across overlapping scanning routes. Multiscale feature extraction provides a promising solution to this challenge by capturing spatial dependencies at different levels of granularity. By combining fine-grained local details with broader contextual information, the model can reduce redundancy while preserving the most informative features. Thus, exploring an effective multiscale strategy is critical for improving the efficiency of SSM-based approaches.
2) *The Heterogeneous gap between multisource data:* SSM's role in multisource remote sensing data classification has not yet been fully explored, as SSM lacks a design similar to cross-attention. This prompts us to investigate how to use SSM to alleviate the heterogeneous gap between HSI and LiDAR/SAR data.

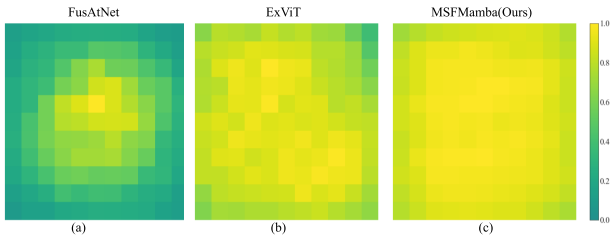


Fig. 1. Effective receptive field of three models. (a) CNN-based FusAtNet, (b) Transformer-based ExViT, and (c) proposed MSFMamba. A broader distribution of bright areas indicates a larger effective receptive filed. The proposed MSFMamba achieves the largest effective receptive field.

To address the above challenges, we propose the Multi-Scale Feature Fusion Mamba (MSFMamba) network for the joint classification of HSI and LiDAR/SAR data. MSFMamba mainly comprises three parts: the Multi-Scale Spatial Mamba (MSpa-Mamba) block, the spectral Mamba (Spe-Mamba) block, and the fusion Mamba (Fus-Mamba) block. Specifically, to solve the feature redundancy in multiple scanning routes, the MSpa-Mamba block incorporates the multiscale strategy to minimize the computational redundancy and alleviate the feature redundancy in SSM. In addition, Spe-Mamba is designed for spectral feature exploration, which is essential for HSI feature modeling. Moreover, to alleviate the heterogeneous gap between HSI and LiDAR/SAR data, we design the Fus-Mamba block for multisource feature fusion. The original Mamba is extended to accommodate dual inputs, and cross-modal feature interaction is enhanced.

In summary, our main contributions can be summarized as follows.

1) We propose MSFMamba, a simple yet effective sequential scanning model for HSI and LiDAR/SAR data interpretation, which investigates the use of SSMs for multisource remote sensing image classification.
2) We design an MSpa-Mamba block that incorporates a multiscale strategy to minimize computational burden, along with a Fus-Mamba block to enhance cross-modal feature interactions between multisource data.
3) Extensive experiments on four public benchmark datasets demonstrate that our method outperforms state-of-the-art techniques in multisource remote sensing image classification.

The remainder of this article is organized as follows. Section II reviews the related work. Section III introduces the preliminaries of the state space model (SSM) and the detailed structure of MSFMamba. Experiments on four multisource datasets are presented in Section IV. A conclusion is drawn, and future work is discussed in Section V.

II. Related Works

A. Deep Learning-Based Multisource Image Classification

Deep learning-based remote sensing image analysis has witnessed significant progress recently [45], [46], [47], [48]. Convolutional neural networks (CNNs) have been widely used in an end-to-end manner for multisource feature fusion.

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Dong et al. [31] used self-attention and cross-guided attention to emphasize regions and channels of interest. Gao et al. [34] presented a cross-mixing attention network to extract multiscale features for hyperspectral and multispectral image joint classification. Xu et al. [49] introduced a dual-tunnel CNN architecture that separately extracts spectral–spatial features from HSI and LiDAR data. Zhang et al. [50] proposed a model known as patch-to-patch CNN for multisource data classification. They constructed a three-tower structure to compute multiscale features. Mohla et al. [51] proposed Feature Fusion And exTraction Network (FusAtNet), which leverages HSI to generate an attention map via self-attention mechanism, which effectively highlights the intrinsic spectral features. In addition, cross-attention is employed to fuse the multisource features. Fang et al. [52] presented a spatial–spectral enhancement module to promote multimodal feature interaction.

To enhance the long-range feature dependencies, many Transformer-based methods have been proposed for multisource remote sensing image classification. Hu et al. [53] proposed a simple yet effective parallel Transformer model. One Transformer acts as an HSI feature encoder, and the other one is responsible for cross-modal feature interactions. Zhao et al. [54] proposed a CNN and Transformer hybrid model for HSI and LiDAR data classification. Cross-token attention is used for cross-modal feature fusion. Yao et al. [55] proposed an Extended Vision Transformer (ExViT), which modifies the traditional Vision Transformer to handle land use and land cover classification tasks. This framework processes multimodal remote sensing image data through parallel branches and introduces a cross-modality attention module to enhance information exchange between different modalities. Li et al. [56] introduced mixing attention and convolution network for multisource feature fusion. Although these CNN and Transformer-based methods have gained performance improvement, there is still an urgent need for an efficient and effective scheme for heterogeneous representation. In this article, we explore Mamba's potential for multisource remote sensing feature fusion and provide an effective paradigm to propel the progress of heterogeneous feature fusion.

B. State Space Model

SSMs [57], [58] have become practical components for constructing deep networks due to their cutting-edge performance in analyzing continuous long sequential data. Gu et al. [58] introduced a diagonal structure and combined it with a diagonal plus low-rank approach to construct structured SSM. Smith et al. [40] improved the efficiency of SSM by introducing parallel scanning techniques. Mamba [41] incorporates data-dependent parameters to amend the linear time-invariant characteristics of SSM-based models, and it has demonstrated excellent performance over Transformers on large-scale datasets.

Recently, inspired by pioneering SSMs, Mamba-like frameworks have been employed in computer vision and remote sensing image interpretation tasks, such as Visual SSM (VMamba) [59], remote sensing images semantic segmentation Mamba (RS^{3}Mamba) [60], and Remote Sensing Mamba (RSMamba) [61]. RSMamba [61] employed a dynamic multipath activation mechanism to enhance the performance of remote sensing image classification. Zhu et al. [62] employed a unique encoder architecture, based on the Mamba design, to effectively extract semantic information from remote sensing images. Chen et al. [63] used a Mamba-like feature encoder to learn global spatial contextual information for remote sensing image change detection. Li et al. [64] proposed an Spe-Mamba block to extract the spectral features of HSI and improve the HSI classification performance. These works show the potential for Mamba's extension into multisource data fusion, but its particular application in the joint classification task of HSI and LiDAR/SAR data remains unexplored.

III. Methodology

In this section, we start by presenting the essential concepts of the SSM. Following that, we delve into a comprehensive description of our MSFMamba, covering its framework and module design in detail.

A. Preliminaries

1) State Space Model: SSM is a linear time-invariant system that maps an input sequence $x(t) \in \mathbb{R}^N$ to an output sequence $y(t) \in \mathbb{R}^N$. They are mathematically represented by linear ordinary differential equations

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t) \tag{2}$$

where $h(t) \in \mathbb{R}^N$ indicates a hidden state, $h'(t) \in \mathbb{R}^N$ refers to the time derivative of $h(t)$, and N represents the number of states. In addition, $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition matrix, $\mathbf{B} \in \mathbb{R}^{N\times 1}$ and $\mathbf{C} \in \mathbb{R}^{N\times 1}$ are projection matrices, and $\mathbf{D} \in \mathbb{R}^{N\times 1}$ is served as a residual connected operation.

SSMs are continuous-time models, and it is challenging for them to be incorporated into deep learning networks. To solve the problem, discrete versions of SSMs are proposed, and the ordinary differential equations can be discretized by the zeroth-order hold rule. A timescale parameter Δ is incorporated to convert the continuous parameters $\mathbf{A}$, $\mathbf{B}$ into discrete parameters $\overline{\mathbf{A}}$, $\overline{\mathbf{B}}$, respectively,

$$\overline{\mathbf{A}} = \exp(\Delta\mathbf{A}) \tag{3}$$

$$\overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \boldsymbol{I}) \cdot \Delta\mathbf{B} \approx \Delta\mathbf{B}. \tag{4}$$

In practice, as noted in [41], the projection matrix $\overline{\mathbf{B}}$ can be approximated by applying a first-order Taylor expansion to the term involving the matrix exponential. After discretization, the ordinary differential equations of SSMs can be represented as follows:

$$h_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t$$
$$y_t = \mathbf{C}h_t + \mathbf{D}x_t. \tag{5}$$

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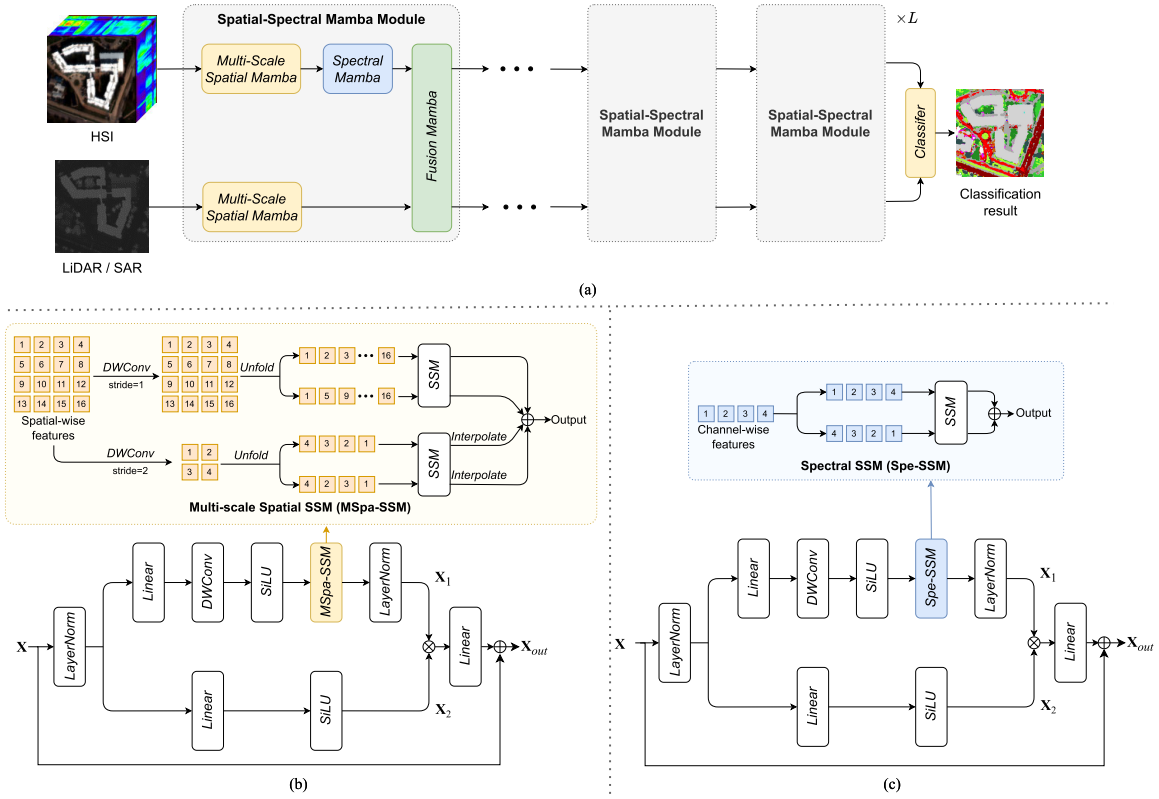


Fig. 2. Framework of the proposed MSFMamba for HSI and LiDAR/SAR data classification. (a) Details of the proposed MSFMamba. It contains L spatial-Spe-Mamba module to learn multisource feature representations. Each spatial-Spe-Mamba module is comprised of the MSpa-Mamba block, the Spe-Mamba block, and the Fus-Mamba block. (b) Details of the MSpa-Mamba block and the MSpa-SSM. (c) Details of the Spe-Mamba block and the Spe-SSM.

2) Selective Scan Mechanism: Traditional SSMs use a linear time-invariant framework, which means that the projection matrices remain fixed and unaffected by variations in the input sequence. However, this static configuration results in a lack of attention to individual elements within the sequence. To alleviate this limitation, Mamba [41] proposes a solution where the parameter matrices become input-dependent. In this way, SSMs can better manage complex sequences, potentially enhancing their capability through the transformation into linear time-varying systems.

B. Overall Framework of the Proposed MSFMamba

As depicted in Fig. 2, our MSFMamba consists of two parts: 1) the main network with a series of L spatial-Spe-Mamba modules to learn high-quality multisource feature representations and 2) a classifier with two fully connected layers for land cover classification. The number of spatial-Spe-Mamba module L is a critical parameter that will be discussed in the experiments.

The HSI is first handled by principal component analysis (PCA) to select the best N_p spectral bands. The HSI and LiDAR/SAR data are fed into the spatial-Spe-Mamba modules for feature extraction and cross-modal feature fusion. The spatial-Spe-Mamba modules are repeated L times, and the final features are concatenated and fed into the classifier for classification. The classifier consists of two fully connected layers.

The Spatial-Spe-Mamba Module is the critical component in our MSFMamba. Details of the Spatial-Spe-Mamba Module are shown in Fig. 2(a). The input HSI and LiDAR/SAR features are first fed into the MSpa-Mamba block for spatial feature extraction. To further exploit the spectral features of HSI, the Spe-Mamba block is employed for HSI feature extraction. Then, we obtain optimized features $\mathbf{F}_h$ and $\mathbf{F}_x$. Next, $\mathbf{F}_h$ and $\mathbf{F}_x$ are fed into the Fus-Mamba block for cross-modal feature fusion and refinement. Finally, the refined HSI feature $\mathbf{F}'_h$ and LiDAR/SAR feature $\mathbf{F}'_x$ are generated.

As can be observed from Fig. 2(a), the MSpa-Mamba block, the Spe-Mamba block, and the Fus-Mamba block are three key components in the Spatial-Spe-Mamba Module. We will give detailed descriptions of the three blocks in Sections III-C–III-E.

C. MSpa-Mamba Block

1) Overview of the MSpa-Mamba: The Mamba can model long-range feature dependencies, which is critical for understanding the global context in remote sensing images. Existing SSM-based models commonly use a multiscan strategy to ensure that every part of the image can establish connections with other parts. However, the multiscan strategy significantly increases the feature redundancy of SSM.

To solve the problem, we design a simple yet effective MSpa-Mamba block. As shown in Fig. 2(b), the input feature

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$\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ is fed into two parallel branches. In the first branch, the feature is processed by a linear layer, followed by a depthwise convolution (DWConv) and a sigmoid linear unit (SiLU) activation function, together with the multiscale spatial SSM (MSpa-SSM) layer and LayerNorm. In the second branch, the feature is processed by a linear layer followed by the SiLU activation function. After that, features from the two branches are aggregated with elementwise multiplication. The computation of the MSpa-Mamba block is given as follows:

$$\mathbf{X}_1 = \text{LN}(\text{MSpa-SSM}(\text{SiLU}(\text{DWConv}(\text{Linear}(\mathbf{X}))))) \quad (6)$$

$$\mathbf{X}_2 = \text{SiLU}(\text{Linear}((\mathbf{X}))) \quad (7)$$

$$\mathbf{X}_{\text{out}} = \text{Linear}(\mathbf{X}_1 \odot \mathbf{X}_2) \quad (8)$$

where LN denotes the LayerNorm and $\odot$ denotes the elementwise multiplication.

2) MSpa-SSM: As shown in Fig. 2(b), MSpa-SSM is the critical part of the MSpa-Mamba block. In MSpa-SSM, features at multiple scales are generated through DWConv with different stride values. These multiscale feature maps are then handled by four scanning routes within SSM [59]. The scanning routes are divided into two groups: two maintain the original resolution and are processed by the SSM, while the others are downsampled, processed by the SSM, and then upsampled. This strategy reduces the overall feature volume by generating more compact feature representations, effectively preventing the accumulation of redundant information across all scanning routes. To be more specific, we use DWConv with strides of 1 and 2 to obtain feature map $\mathbf{Z}_1 \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{Z}_2 \in \mathbb{R}^{(H/2)\times(W/2)\times C}$, respectively. Next, $\mathbf{Z}_1$ and $\mathbf{Z_2}$ are fed into the SSM as follows:

$$[\mathbf{Y}_1, \mathbf{Y}_2] = \text{SSM}(\sigma_1(\mathbf{Z}_1)\sigma_2(\mathbf{Z}_1)) \quad (9)$$

$$[\mathbf{Y}_3, \mathbf{Y}_4] = \text{SSM}(\sigma_3(\mathbf{Z}_2)\sigma_4(\mathbf{Z}_2)) \quad (10)$$

where σ represents the transformation that reshapes the input features from $H \times W \times C$ to $HW \times C$, where HW specifies the length of the spatial sequence to be processed by the SSM module. As illustrated in Fig. 2, the input is scanned in four distinct directions: the first scans rows first and then columns, while the second scans columns first and then rows. The third and fourth directions are the reverse of the first and second, respectively. Consequently, $\mathbf{Y}$ denotes the resulting processed sequence.

The obtained sequences are converted back into 2-D feature maps, and the downsampled feature maps are interpolated for merging as follows:

$$\mathbf{Z}'_i = \beta_i(\mathbf{Y}_i), \quad i \in \{1, 2, 3, 4\} \quad (11)$$

$$\mathbf{Z}' = \mathbf{Z}'_1 + \mathbf{Z}'_2 + \text{Inter}(\mathbf{Z}'_3 + \mathbf{Z}'_4) \quad (12)$$

where β is the inverse transformation of σ, reshaping the input data from $\mathbb{R}^{HW\times C}$ back to $\mathbb{R}^{H\times W\times C}$. Here, $\mathbf{Z}'$ represents the feature map enhanced by the SSM module, and "Inter" refers to the interpolation operation.

D. Spe-Mamba Block

HSI covers a much larger spectral range with a higher spectral resolution. Effectively modeling the spectral relationship is important. In this article, we design a Spe-Mamba block for spectral feature modeling.

Details of the Spe-Mamba are illustrated in Fig. 2(c). The input feature is fed into two parallel branches. In the first branch, the feature is handled by a linear layer, DWConv, and SiLU activation, together with the Spectral SSM (Spe-SSM) layer. In the second branch, the feature is handled by a linear layer and SiLU activation. Finally, features from both branches are aggregated with elementwise multiplication.

The Spe-Mamba shares a similar structure to the MSpa-Mamba, with the primary distinction being the SSM layer. In MSpa-SSM, features are transformed into 1-D sequences along the spatial dimension, while in Spe-Mamba, they are converted into 1-D sequences along the channel dimension, with scanning limited to two directions. Before being fed into Spe-SSM, the feature dimensions are reshaped from $\mathbb{R}^{H\times W\times C}$ to $\mathbb{R}^{C\times HW}$, where C specifies the length of the sequence processed by the SSM module.

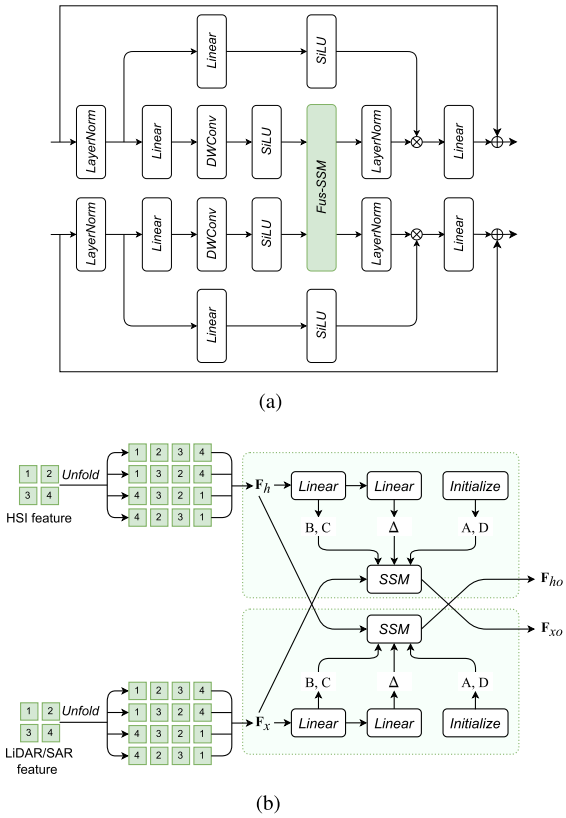


Fig. 3. Illustration of (a) Fus-Mamba block and (b) Fus-SSM.

E. Fus-Mamba Block

Traditional SSMs typically handle single inputs and struggle to effectively integrate cross-modal information from multisource data. To address this limitation, we propose the Fus-Mamba block, which integrates features from different modalities. As shown in Fig. 3, HSI and LiDAR/SAR features undergo a similar processing flow to that of MSpa-Mamba and Spe-Mamba. Multisource features are then fed into the Fusion-SSM (Fus-SSM) block for cross-modal feature fusion and interaction.

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Algorithm 1 Computation of $\mathbf{F}_{ho}$ in Fig. 3(b)

Input: $\mathbf{F}_h, \mathbf{F}_x$: (P, C), (P, C)
Output: $\mathbf{F}_{ho}$: (P, C)

$\mathbf{A}$: (C, N) $\leftarrow$ **Parameter**
$\mathbf{D}$: (C) $\leftarrow$ **Parameter**
/* A and D are randomly initialized */
$\mathbf{B}$: (P, N) $\leftarrow$ **Linear**($\mathbf{F}_x$)
$\mathbf{C}$: (P, N) $\leftarrow$ **Linear**($\mathbf{F}_x$)
Δ : (P, C) $\leftarrow \log(1 + \exp(\textbf{Linear}(\mathbf{F}_x) + \textbf{Parameter})$
/* The selection mechanism */
$\overline{\mathbf{A}}$: (P, C, N) $\leftarrow \exp(\Delta \otimes \mathbf{A})$
$\overline{\mathbf{B}}$: (P, C, N) $\leftarrow \Delta \otimes \mathbf{B}$
$\mathbf{F}_{ho} \leftarrow \text{SSM}(\overline{\mathbf{A}}, \overline{\mathbf{B}}, \mathbf{C}, \mathbf{D})(\mathbf{F}_h)$
/* SSM denotes Eq. 5 implemented by selective scan */
return $\mathbf{F}_{ho}$

The Fus-SSM serves as the core of Fus-Mamba, with its structure detailed in Fig. 3(b). Fus-SSM is designed symmetrically. In the top branch, the sequenced HSI feature $\mathbf{F}_h \in \mathbb{R}^{P\times C}$ generates projection and timescale parameters, while the sequenced LiDAR/SAR feature $\mathbf{F}_x \in \mathbb{R}^{P\times C}$ is processed to produce the output feature $\mathbf{F}_{xo}$. The bottom branch mirrors this structure: here, $\mathbf{F}_x$ generates the projection and timescale parameters, and $\mathbf{F}_h$ is processed to generate the output feature $\mathbf{F}_{ho}$.

Algorithm 1 outlines the computation process for $\mathbf{F}_{ho}$. Parameters A and D are initialized at the start, while B, C, and Δ are derived from linear layers applied to the sequence $\mathbf{F}_x$ from another modality. In the original Mamba model, B, C, and Δ are derived from the sequence being processed, whereas in Fus-SSM, B, C, and Δ come from a different modality, specifically $\mathbf{F}_x$. Since the computation for $\mathbf{F}_{xo}$ closely resembles that of $\mathbf{F}_{ho}$, it is omitted for brevity.

Comparison with Cross Mamba Block (CroMB): Although the Fus-Mamba Block shares conceptual similarities with the CroMB in Sigma [65], it incorporates a more integrated cross-modal design. In CroMB, components $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ are generated solely from the sequence being processed, with only $\mathbf{C_X}$ relying on the sequence from the alternate modality. In contrast, our Fus-SSM utilizes features from the alternative modality to generate all three parameters ($\overline{\mathbf{A}}$, $\overline{\mathbf{B}}$, and $\mathbf{C}$) when processing a given sequence. This approach enables the Fus-Mamba Block to capture deeper interactions and dependencies between modalities and fully leverage cross-modal information to improve feature alignment and enhance overall model performance.

IV. Experimental Results and Analysis

A. Dataset Description

To assess the effectiveness of our proposed MSFMamba, we applied it to four multisource remote sensing datasets: Berlin, Augsburg, Houston 2018, and Houston 2013. The Berlin and Augsburg datasets are used for hyperspectral and SAR data classification, while the Houston 2013 and Houston 2018 datasets are used for hyperspectral and LiDAR data classification.

TABLE I
Number of Training and Test Samples for the Berlin, Augsburg, Houston2018, and Houston2013 Datasets

No.	Name	Color	Train	Test
1	Forest		443	54511
2	Residential area		423	268219
3	Industrial area		499	19067
4	Low plants		376	58906
5	Soil		331	17095
6	Allotment		280	13025
7	Commercial area		298	24526
8	Water		170	6502
	Total		2820	461851

Augsburg dataset

No.	Name	Color	Train	Test
1	Forest		146	13361
2	Residential area		264	30065
3	Industrial area		21	3830
4	Low plants		248	26609
5	Allotment		52	523
6	Commercial area		7	1638
7	Water		23	1507
	Total		761	77533

Houston2018 dataset

No.	Name	Color	Train	Test
1	Health grass		1000	38196
2	Stressed grass		1000	129008
3	Artificial turf		1000	1736
4	Evergreen trees		1000	53322
5	Deciduous trees		1000	19172
6	Bare earth		1000	17064
7	Water		500	564
8	Residential buildings		1000	157995
9	Non-residential buildings		1000	893769
10	Roads		1000	182283
11	Sidewalks		1000	135035
12	Crosswalks		1000	5059
13	Major thoroughfares		1000	184438
14	Highways		1000	38438
15	Railways		1000	26748
16	Paved parking lots		1000	44932
17	Unpaved parking lots		250	337
18	Cars		1000	25289
19	Trains		1000	20479
20	Stadium seats		1000	26296
	Total		18750	2000160

Houston2013 dataset

No.	Name	Color	Train	Test
1	Health grass		198	1053
2	Stressed grass		190	1064
3	Synthetic grass		192	505
4	Trees		188	1056
5	Soil		186	1056
6	Water		182	143
7	Residential		196	1072
8	Commercial		191	1053
9	Road		193	1059
10	Highway		191	1036
11	Railway		181	1054
12	Parking lot 1		192	1041
13	Parking lot 2		184	285
14	Tennis court		181	247
15	Running track		187	473
	Total		2832	12197

1) Berlin Dataset: It covers the urban and rural areas of Berlin. The hyperspectral data are simulated Environmental Mapping and Analysis Program (EnMAP) data based on Hyperspectral Mapper (HyMap) data. The SAR data are

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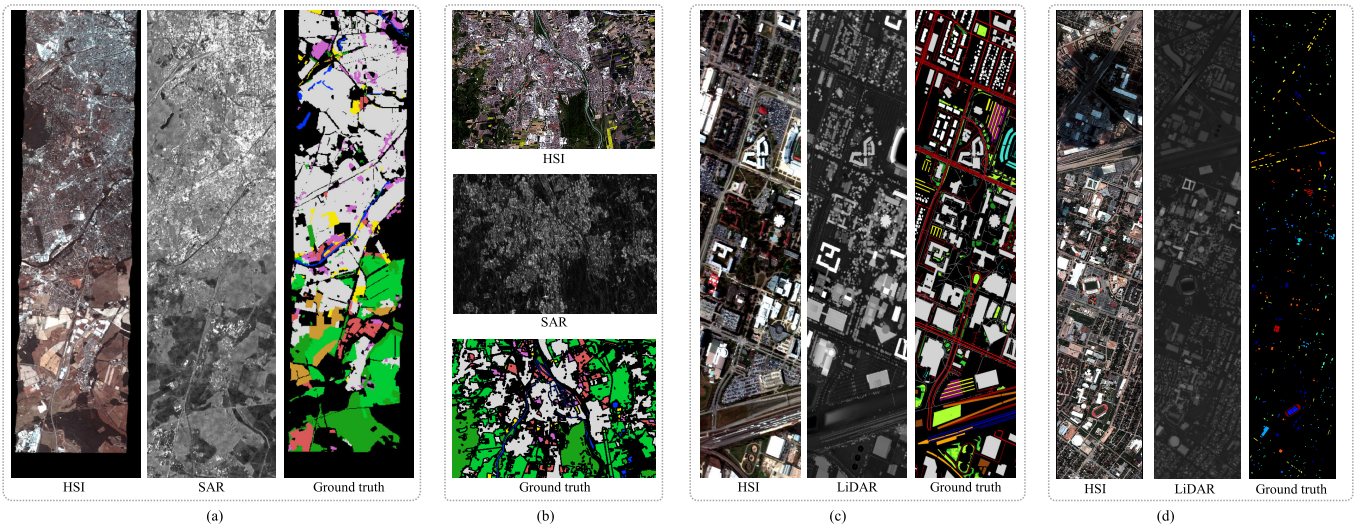


Fig. 4. Visualization of the four datasets used in our experiments. (a) Berlin dataset. (b) Augsburg dataset. (c) Houston2018 dataset. (d) Houston2013 dataset.

Sentinel-1 dual-Pol (VV–VH) single-look complex (SLC) product obtained from the European Space Agency (ESA) [66]. The HSI has 797 × 220 pixels and 244 spectral bands in the wavelength range of 400–2500 nm [67]. The SAR image has 1723 × 476 pixels. The nearest neighbor interpolation was employed to match the spatial resolution of the HSI and SAR data.

2) Augsburg Dataset: This dataset is composed of HSI and SAR data from Augsburg, Germany. The HSI was acquired by the HySpex sensor, and the SAR data were acquired by the Sentinel-1 sensor. All images are with a ground sample distance (GSD) of 30 m. The spatial size of both images is 332 × 485 pixels. The HSI contains 180 spectral bands ranging from 0.4 to 2.5 μm, and the SAR data have four features derived from polarization decomposition (VV intensity, VH intensity, real part, and imaginary part of the off-diagonal element of the PolSAR covariance matrix).

3) Houston2018 Dataset: This dataset covers the University of Houston campus and the neighboring urban area. The HSI contains 48 bands in the wavelength range of 380–1050 nm. LiDAR data are multispectral LiDAR data with three bands. The University of Houston released the dataset as part of the 2018 IEEE GRSS Data Fusion Contest [68]. We use the training subset of the whole dataset in this article.

4) Houston2013 Dataset: The Houston2013 dataset, part of the 2013 IEEE Geoscience and Remote Sensing Society Data Fusion Contest, provides a unique perspective on urban land cover in Houston, Texas, and its surrounding areas. Captured at a spatial resolution of 2.5 m, the dataset includes hyperspectral imagery with 144 spectral bands covering wavelengths from 380 to 1050 nm, along with LiDAR data that provide precise elevation information. Together, these data sources encompass 15 distinct land cover types, offering valuable insights for remote sensing and urban analysis.

Table I enumerates the number of samples for both training and testing on the four datasets. Fig. 4 displays the HSI data in false color, along with the LiDAR/SAR images and the ground truth. The evaluation metrics used in this article include

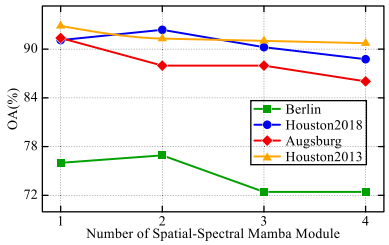


Fig. 5. Relationship between OA and the number of spatial-Spe-Mamba modules.

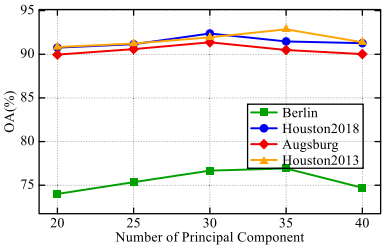


Fig. 6. Relationship between OA and the number of principal components in HSI.

overall accuracy (OA), average accuracy (AA), and kappa. OA represents the proportion of correctly classified samples out of all the samples in the dataset, providing a measure of the model's overall performance. AA is the average recall rate across all classes, reflecting the model's effectiveness in handling different categories, particularly in imbalanced datasets. Kappa, on the other hand, is a statistical measure that quantifies the agreement between the predicted and true labels, adjusted for random chance, offering a more robust evaluation of classifier performance than simple accuracy, especially when dealing with imbalanced or noisy data.

B. Parameter Analysis

We provide a detailed analysis of several important parameters that may affect the classification performance of our MSFMamba. These parameters include the number

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TABLE II
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE BERLIN DATASET

Class	TBCNN	FusAtNet	S^2ENet	DFINet	AsyFFNet	ExViT	HCT	MACN	MSFMamba
Forest	81.75	86.24	83.27	82.04	**88.35**	84.71	83.18	82.21	82.39
Residential area	76.26	**91.38**	72.07	77.78	74.44	76.19	78.80	78.16	82.05
Industrial area	39.67	19.76	46.66	47.96	48.21	44.15	43.59	47.79	**48.91**
Low plants	49.78	20.00	72.08	77.78	71.41	77.96	76.48	78.04	**79.72**
Soil	**89.42**	48.72	77.94	87.85	80.75	67.13	82.92	77.26	85.22
Allotment	54.36	38.89	70.62	44.75	48.35	61.95	56.46	66.07	**71.25**
Commercial area	4.65	18.47	**36.48**	29.85	27.97	29.28	18.96	28.90	31.08
Water	41.93	29.61	54.64	60.09	**75.16**	56.98	44.16	50.22	38.48
OA	67.60	70.91	70.38	73.69	71.65	72.59	73.42	73.98	**76.92**
AA	54.72	44.13	64.22	63.51	64.33	62.39	60.57	63.58	**64.88**
Kappa	50.96	51.07	57.73	61.02	58.74	59.72	60.29	61.19	**64.88**

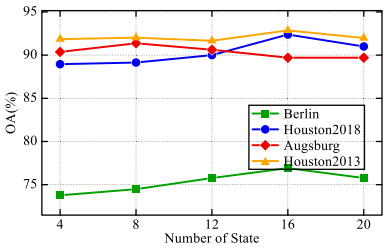


Fig. 7. Relationship between OA and the number of states in SSM.

of spatial-Spe-Mamba modules, the number of principal components in HSI, and the number of states in SSM.

1) Number of Spatial-Spe-Mamba Module: The spatial-Spe-Mamba module is the critical part of our MSFMamba, and the number of modules, L, is a key parameter that affects classification performance. We tested the optimal number of spatial-Spe-Mamba modules for each dataset. Fig. 5 illustrates the relationship between the OA value and L on each dataset. Our MSFMamba achieves the best classification performance on the Augsburg and Houston2013 datasets when $L = 1$. For the Berlin and Houston2018 datasets, the best performance is observed when $L = 2$. Therefore, in our subsequent experiments, we set $L = 1$ for the Augsburg and Houston2013 datasets and $L = 2$ for the Berlin and Houston2018 datasets.

2) Number of Principal Components in HSI: We use PCA to reduce the spectral redundancy of the input HSI, with the number of principal components, N_p, being a critical parameter. We tested different values of N_p ranging from 20 to 40, and the experimental results are shown in Fig. 6. It can be seen that when $N_p = 30$, the OA values for the Augsburg and Houston2018 datasets reach their highest. For the Berlin and Houston2013 datasets, the best OA values are achieved when $N_p = 35$.

3) Number of States in SSM: The number of states is an important parameter in the SSM. We tested five different values for the number of states, ranging from 4 to 20. As shown in Fig. 7, the optimal number of states for the Houston2018, Houston2013, and Berlin datasets is 16, while, for the Augsburg dataset, the optimal number of states is 8.

The best settings of these experiments were used to evaluate our MSFMamba for comparison with other state-of-the-art methods. All experiments were performed with an RTX 4090 GPU and 32 GB of RAM using PyTorch.

C. Performance Comparison

We compared the proposed MSFMamba with existing state-of-the-art methods, including the two-branch CNN (TBCNN) [49], the FusAtNet, the spatial–spectral cross-modal enhancement network (S^2ENet) [52], the depthwise feature interaction network (DFINet) [69], the asymmetric feature fusion network (AsyFFNet) [70], ExViT, the hierarchical CNN and Transformer (HCT) [54], and the mixing self-attention and convolution network (MACN) [56]. TBCNN [49] investigates the classification fusion of hyperspectral imagery and data from other sensors using a two-branch convolution neural network. FusAtNet [51] presents a multisource classification framework by utilizing a self-attention mechanism for spectral features and a cross-attention approach for spatial features. S^{2}ENet [52] proposes a spatial–spectral enhancement module for cross-modal information interaction, which effectively facilitates the feature interaction between multisource data. DFINet [69] uses a depthwise cross-attention module to extract complementary information from multisource feature pairs. AsyFFNet [70] is a multisource data classification method based on asymmetric feature fusion, which employs weight-share residual blocks for feature extraction and a feature calibration module for the spatialwise multisource feature modeling. ExViT [55] utilizes parallel branches of position-shared Transformer extended with separable convolution modules to process multimodal image patches. In MACN [56], mixing self-attention and convolution Transformer layer is proposed to extract local and global multiscale feature perception. The classification performance of various methods on multiple datasets is presented in Tables II–V. These tables compare the recall rates of different methods for specific classes, providing a detailed evaluation of their performance.

1) Results on the Berlin Dataset: The classification performance of various methods on the Berlin dataset is presented in Table II, with corresponding classification maps shown in Fig. 8. Our proposed MSFMamba achieves the highest OA of 76.92%, an AA of 64.88%, and a kappa coefficient of 64.88%. These results highlight the model's robust multiscale spatial contextual modeling capabilities, enabled by its sequential modeling approach. A closer analysis reveals that MSFMamba excels in distinguishing fine-grained classes, particularly industrial areas, low plants, and allotment, where capturing intricate textures and boundary details is

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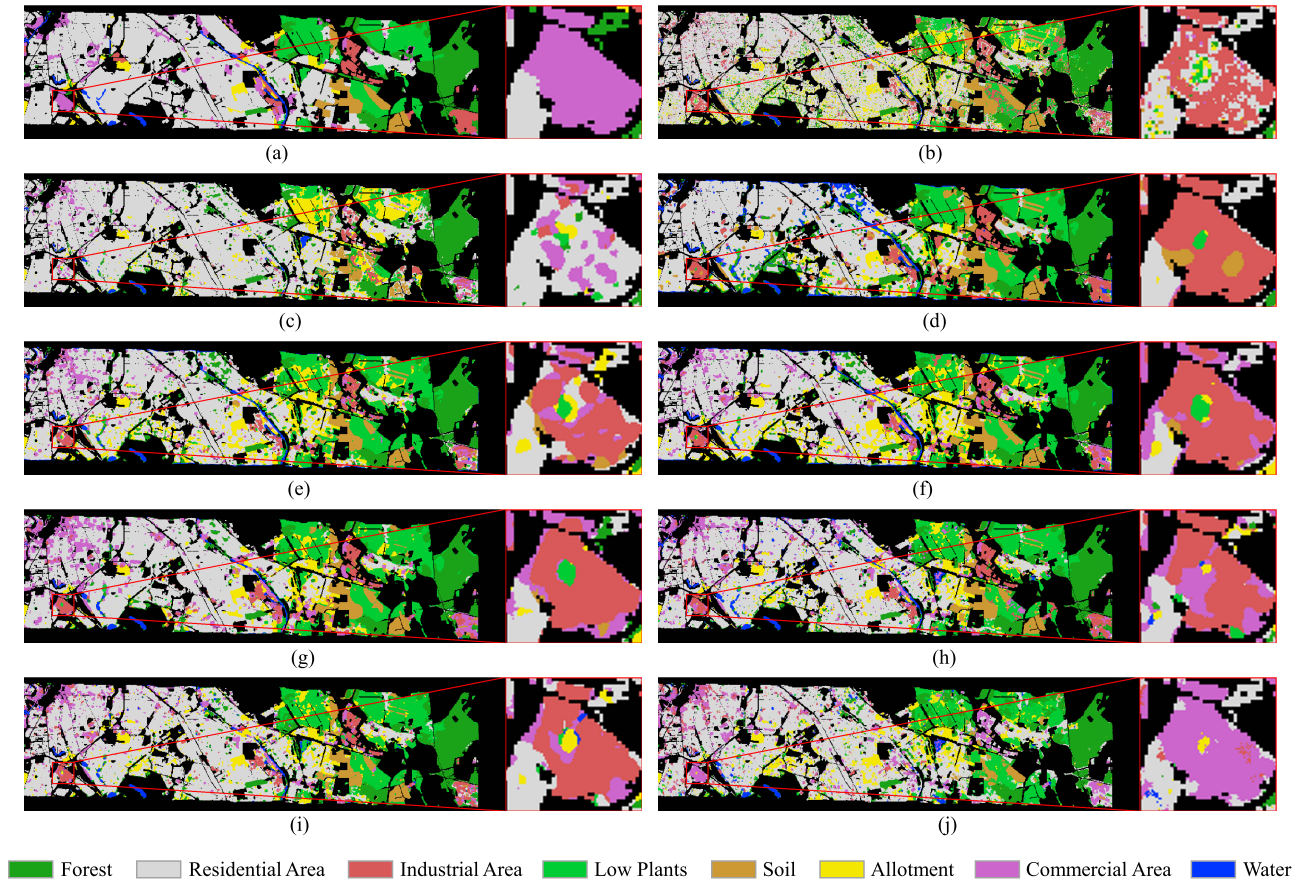


Fig. 8. Classification results of different methods for the Berlin dataset. (a) Ground truth. (b) TBCNN. (c) FusAtNet. (d) S^2ENet. (e) DFINet. (f) AsyFFNet. (g) ExViT. (h) HCT. (i) MACN. (j) MSFMamba (Ours).

TABLE III
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE AUGSBURG DATASET

Class	TBCNN	FusAtNet	S^2ENet	DFINet	AsyFFNet	ExViT	HCT	MACN	MSFMamba
Forest	90.88	94.34	**98.10**	97.38	97.56	90.04	97.26	97.84	97.17
Residential area	93.89	92.56	99.08	98.37	**99.16**	95.44	98.63	98.73	98.15
Industrial area	8.28	47.70	12.19	61.31	**61.38**	34.58	36.61	43.94	50.26
Low plants	91.97	85.96	91.78	92.63	83.47	90.68	93.54	94.11	**95.52**
Allotment	38.24	49.33	45.12	49.33	42.64	51.82	51.43	52.39	**53.35**
Commercial area	1.40	11.52	1.22	3.54	5.58	**28.63**	7.97	6.80	2.63
Water	10.82	45.27	24.09	26.61	45.81	17.65	46.34	48.54	**49.97**
OA	84.53	85.31	88.22	90.66	88.25	86.65	90.33	91.06	**91.38**
AA	47.93	62.93	53.08	61.31	62.23	58.41	61.68	63.19	**63.31**
Kappa	77.13	86.33	86.47	86.47	83.13	80.79	86.03	87.06	**87.45**

critical. The model effectively handles these challenges, resulting in more accurate classification. In the visualized results, particularly in the magnified sections, it is evident that MSFMamba produces classification maps much closer to the ground truth. While other methods tend to misclassify areas such as commercial areas, MSFMamba more accurately captures these regions, showcasing its ability to preserve finer details and textures in challenging areas.

2) Results on the Augsburg Dataset: Table III presents the quantitative results of various methods on the Augsburg dataset. A closer analysis of classwise performance reveals that MSFMamba achieves notable improvements in categories such as low plants, allotment, and water, which are characterized by complex spectral–spatial patterns. Our MSFMamba outperforms the second-best algorithm by 0.22% in OA, 0.12% in AA, and 0.39% in kappa. The corresponding classification maps, as shown in Fig. 9, further demonstrate the advantage of MSFMamba. By effectively extracting spatial–spectral semantic features through its sequential modeling mechanism, MSFMamba produces clearer boundaries and significantly reduces misclassified regions compared to other methods.

3) Results on the Houston2018 Dataset: Table IV presents the classification performance of various methods on the Houston2018 dataset, with Fig. 10 showing the corresponding classification maps. MSFMamba delivers a highly refined classification map, with most noisy regions significantly reduced, resulting in cleaner and more accurate segmentation. On the Houston2018 dataset, MSFMamba demonstrates exceptional

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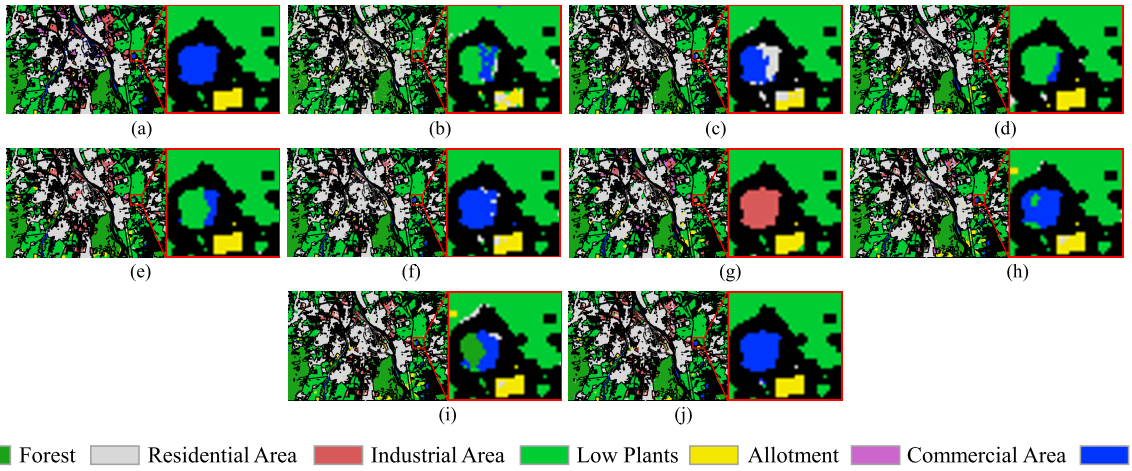


Fig. 9. Classification results of different methods for the Augsburg dataset. (a) Ground truth. (b) TBCNN. (c) FusAtNet. (d) S^2ENet. (e) DFINet. (f) AsyFFNet. (g) ExViT. (h) HCT. (i) MACN. (j) MSFMamba (Ours).

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE HOUSTON 2018 DATASET

Class	TBCNN	FusAtNet	S^2ENet	DFINet	AsyFFNet	ExViT	HCT	MACN	MSFMamba
Health grass	94.84	60.69	**99.15**	94.29	96.9	93.34	97.38	89.53	90.38
Stressed grass	92.6	93.85	84.35	92.71	91.82	93.44	89.4	93.5	**94.04**
Artificial turf	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Evergreen trees	98.8	99.02	98.13	98.92	99.29	95.87	98.9	99.1	**99.34**
Deciduous trees	97.12	94.73	86.39	97.77	**99.52**	96.85	98.89	98.49	99.11
Bare earth	99.61	99.87	99.33	**100.0**	99.98	**100.0**	**100.0**	99.96	**100.0**
Water	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Residential buildings	93.21	95.51	97.17	97.36	95.16	**97.39**	95.3	95.31	96.27
Non-residential buildings	91.3	93.64	93.93	93.29	95.02	93	95.25	93.86	**95.58**
Roads	61.06	70.94	70.17	75.73	72.72	72.18	72.77	72.11	**75.9**
Sidewalks	75.91	79.11	72.11	**83.67**	77.18	69.69	73.27	80.69	81.43
Crosswalks	85.31	86.06	82.47	94.23	93.14	88.12	**98.95**	97.69	96.22
Major thoroughfares	72.77	76.12	**89.44**	81.2	84.67	86.12	80.9	79.05	86.89
Highways	95.86	98.89	98.53	98.91	**99.53**	99.5	98.01	96.86	98.63
Railways	99.78	99.88	99.21	**99.94**	99.90	99.87	99.74	99.9	99.69
Paved parking lots	90.74	95.88	95.62	98.37	**99.18**	97.02	97.07	97.74	98.94
Unpaved parking lots	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Cars	98.47	94.25	96.77	**99.09**	96.64	98.13	99.01	98.74	97.81
Trains	**99.9**	99.84	**100.0**	99.46	**100.0**	99.98	99.97	99.74	99.97
Stadium seats	99.92	**100.0**	**100.0**	99.98	99.89	99.99	99.98	99.99	**100.0**
OA	86.95	89.09	90.05	91.02	91.24	89.98	90.56	90.31	**92.38**
AA	92.36	91.91	93.14	95.24	95.02	94.02	94.74	94.61	**95.51**
Kappa	83.39	85.96	87.2	88.46	88.71	87.14	87.81	87.55	**90.16**

TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE HOUSTON 2013 DATASET

Class	TBCNN	FusAtNet	S^2ENet	DFINet	AsyFFNet	ExViT	HCT	MACN	MSFMamba
Health grass	82.72	82.53	81.48	**82**.91	82.72	82.72	81.1	82.90	82.72
Stressed grass	83.18	85.15	84.68	85.15	83.93	85.15	86.09	**100.0**	99.44
Synthetic grass	**100.0**	98.81	99.01	99.60	**100.0**	99.21	99.80	99.80	98.81
Trees	94.03	92.90	92.14	92.33	91.95	91.95	98.01	98.11	**98.39**
Soil	99.15	**100.0**	**100.0**	99.81	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Water	99.30	98.60	**100.0**	**100.0**	95.80	98.60	95.10	95.80	95.80
Residential	79.66	85.45	89.93	93.47	**96.18**	86.66	83.49	78.17	82.18
Commercial	55.27	81.81	90.18	82.58	81.14	91.05	86.43	80.46	**94.23**
Road	75.83	83.76	92.63	89.05	86.12	91.12	86.69	90.93	**93.96**
Highway	62.55	53.67	64.86	56.56	64.58	64.86	79.05	55.88	**79.92**
Railway	96.87	79.65	97.60	94.15	86.76	77.06	95.97	**99.33**	93.47
Parking lot 1	86.55	91.26	88.09	94.72	90.59	88.76	98.46	**99.71**	96.73
Parking lot 2	53.68	81.05	91.93	89.12	**92.28**	87.72	90.88	88.77	90.88
Tennis court	98.79	**100.0**	**100.0**	**100.0**	95.14	97.17	**100.0**	**100.0**	**100.0**
Running track	98.10	98.73	**1**00.0	**1**00.0	**1**00.0	98.52	**1**00.0	**1**00.0	**1**00.0
OA	82.91	85.32	89.56	88.59	87.95	87.42	90.66	89.81	**92**.86
AA	84.38	87.56	91.50	90.63	89.81	89.37	92.07	91.33	**93.77**
Kappa	81.43	84.12	88.69	87.64	86.96	86.38	89.86	88.93	**92.25**

class-specific performance, especially in categories such as stressed grass, evergreen trees, nonresidential buildings, and roads, where it surpasses other methods by better capturing spatial structures and avoiding common misclassifications. The sequential scanning approach employed by MSFMamba proves to be highly effective for HSI and LiDAR data fusion,

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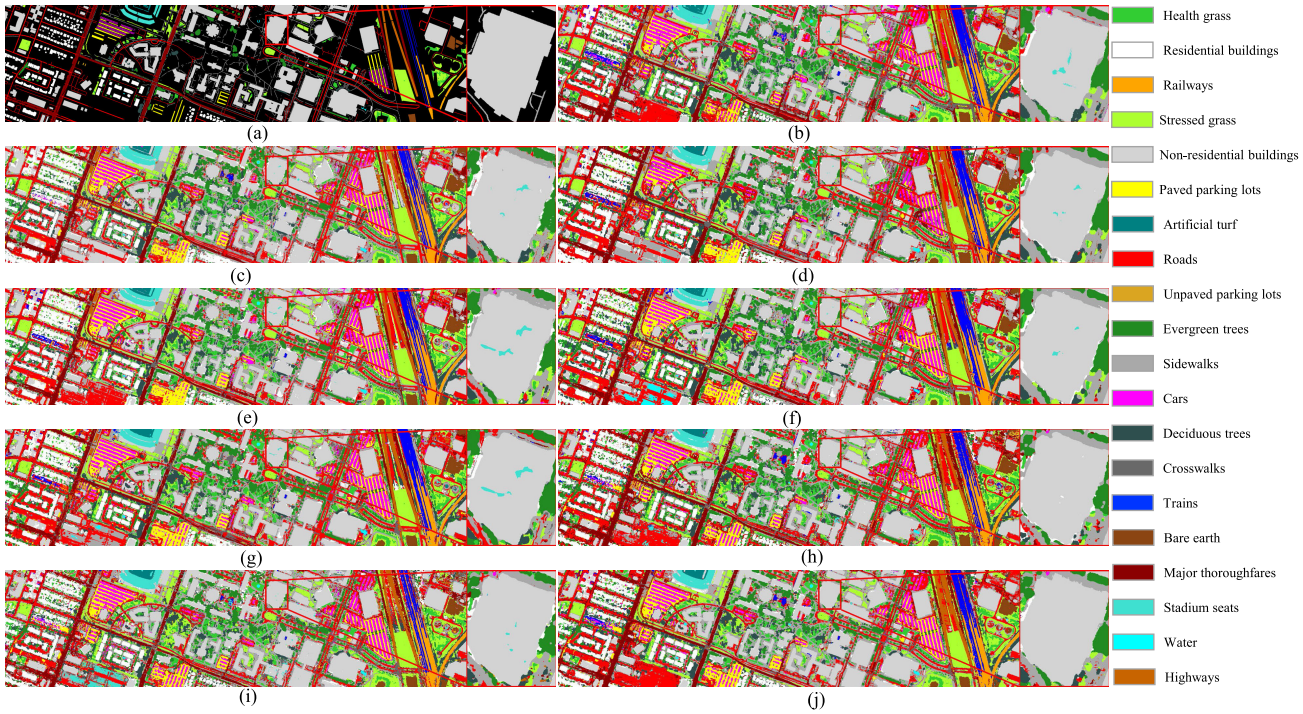


Fig. 10. Classification results of different methods for the Houston 2018 dataset. (a) Ground truth. (b) TBCNN. (c) FusAtNet. (d) S^2ENet. (e) DFINet. (f) AsyFFNet. (g) ExViT. (h) HCT. (i) MACN. (j) MSFMamba (Ours).

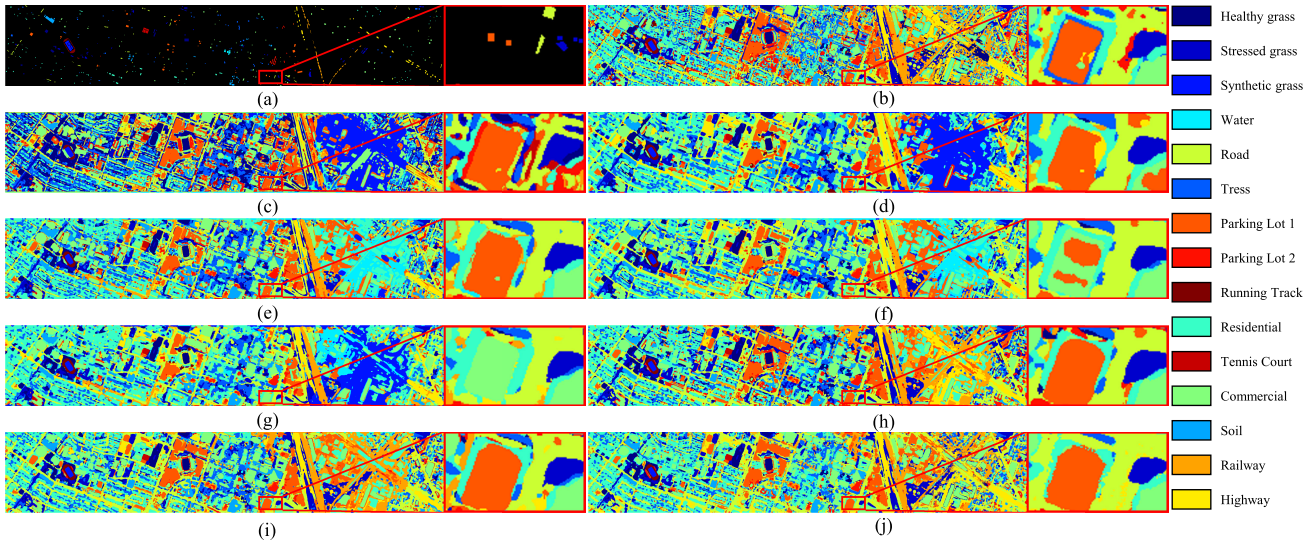


Fig. 11. Classification results of different methods for the Houston 2013 dataset. (a) Ground truth. (b) TBCNN. (c) FusAtNet. (d) S^2ENet. (e) DFINet. (f) AsyFFNet. (g) ExViT. (h) HCT. (i) MACN. (j) MSFMamba (Ours).

leading to more accurate and consistent class predictions, as evidenced by the minimal noise and misclassified areas in the visualized results.

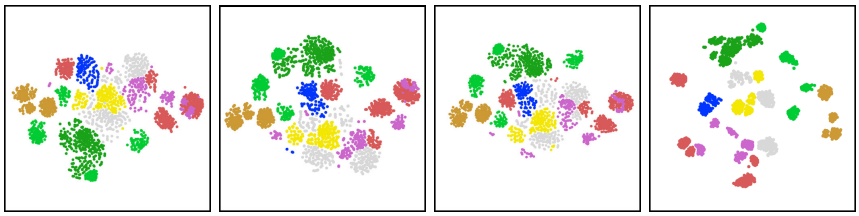
4) Results on the Houston2013 Dataset: Table V summarizes the classification results of various approaches on the Houston2013 dataset, while Fig. 11 illustrates the respective classification maps. MSFMamba produces a well-refined classification output, achieving more precise segmentation and significantly minimizing noise in most areas, resulting in a clearer result overall. On the Houston2013 dataset, MSFMamba exhibits outstanding performance on specific classes, particularly in categories such as trees, commercial, road, and highway. It outperforms other methods by accurately capturing spatial structures and mitigating common misclassification errors.

5) Practical Implications: The experimental results across all datasets highlight MSFMamba's applicability in addressing real-world challenges such as land cover classification and urban infrastructure planning. For example, its high classification accuracy on datasets such as Berlin and Augsburg makes it valuable for urban planning and ecological monitoring, while its robust performance on Houston2018 and Houston2013 datasets supports transportation network analysis and city development projects.

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TABLE VI
INFLUENCE OF MSPA-MAMBA, SPE-MAMBA, AND FUS-MAMBA ON CLASSIFICATION RESULTS OF MSFMAMBA

MSpa-Mamba	Spe-Mamba	Fus-Mamba	Berlin	Augsburg	Houston2018	Houston2013
✔	✖	✖	74.88	89.05	89.56	90.27
✔	✔	✖	76.00	90.60	91.44	91.55
✔	✖	✔	75.88	90.55	91.76	91.82
✔	✔	✔	76.92	91.38	92.38	92.86



(a) (b) (c) (d)

Fig. 12. t-SNE visualization of MSFMamba with different components. (a) MSpa-Mamba. (b) MSpa-Mamba + Spe-Mamba. (c) MSpa-Mamba + Fus-Mamba. (d) Full model of our MSFMamba.

D. Ablation Study

Table VI presents the classification performance of different module combinations on the four datasets. The results reveal a clear trend: the inclusion of each module consistently improves the OA. MSpa-Mamba serves as the fundamental module, achieving OA values of 74.88%, 89.05%, 89.56%, and 90.27% on the four datasets, respectively. Its strong spatial feature extraction capability lays a solid foundation for classification tasks. Adding Spe-Mamba further enhances performance to 76.00%, 90.60%, 91.44%, and 91.55%. This significant improvement underscores the importance of spectral feature extraction in capturing the unique patterns of HSI and SAR/LiDAR data. The combination of MSpa-Mamba and Fus-Mamba yields accuracies of 75.88%, 90.55%, 91.76%, and 91.82%. This demonstrates the critical role of effective multisource data fusion in handling heterogeneous datasets. When all three components are combined, the model achieves the best performance across four datasets, with substantial noise reduction and more precise class boundaries. This demonstrates the complementary nature of the modules: MSpa-Mamba and Spe-Mamba provide robust spatial–spectral feature extraction, while Fus-Mamba effectively integrates these features, maximizing classification accuracy.

We conducted t-distributed stochastic neighbor embedding (t-SNE) visualizations of the three components on the Berlin dataset for our MSFMamba, and the results are illustrated in Fig. 12. t-SNE is a dimensionality reduction technique widely used to visualize high-dimensional data by projecting it into a lower dimensional space while preserving its local structure. It can be observed that with only the MSpa-Mamba, the features of different classes are less compact. With all three modules, the feature representations display the most distinct and well-defined clusters compared to the others. It is evident that the sequential scanning model generates more robust feature representations for multisource data classification.

E. Analysis of Multi-Scale Spatial Mamba

Table VII presents the impact of varying the number of downsampling paths on the classification performance for the Berlin dataset. The results show that the best performance is achieved when two paths retain the original resolution and two paths are downsampled, yielding OA, AA, and kappa values of 76.92%, 64.88%, and 64.88%, respectively. This configuration balances the reduction of redundant features with the preservation of critical details, demonstrating the effectiveness of the multiscale scanning approach. In contrast, configurations with three or more downsampling paths lead to a performance decline, with OA dropping to 75.30% and 74.31%, highlighting the tradeoff between feature reduction and information loss.

This analysis emphasizes the importance of integrating original-resolution and downsampled paths to capture both fine-grained details and broader spatial patterns. The consistent improvement in classification metrics from 0 to 2 downsampling paths underscores the contribution of this multiscale strategy in handling complex spectral–spatial patterns in the Berlin dataset. These results demonstrate that an optimal balance in downsampling is critical for achieving robust performance in remote sensing classification tasks.

TABLE VII
EXPERIMENTAL RESULTS OF THE MULTISCAN STRATEGY ON THE BERLIN DATASET

Path number for downsampling	OA	AA	Kappa
0	74.71	61.15	61.99
1	75.44	62.85	62.93
2	76.92	64.88	64.88
3	75.30	62.50	62.91
4	74.31	61.79	61.99

TABLE VIII
IMPACT OF SCANNING ROUTES ON CLASSIFICATION PERFORMANCE

Scanning Routes	OA(%)	AA(%)	Kappa
2	75.28	63.22	62.28
4	76.92	64.88	64.88
6	74.98	63.45	63.39

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TABLE IX
COMPARATIVE ANALYSIS OF MODEL PARAMETERS, FLOPS, AND INFERENCE TIME ON THE AUGSBURG DATASET

Metrics	FusAtNet	S^2ENet	DFINet	AsyFFNet	ExVit	HCT	MACN	MSFMamba
Params (M)	37.7177	1.4701	1.3155	2.4132	1.8848	4.7811	2.0199	1.5252
FLOPS (G)	3.5619	0.1779	0.1191	0.2064	0.2901	0.0437	0.1672	0.0377
Inference time (s)	0.2469	0.2489	0.3182	0.2924	0.3271	0.7524	0.1917	0.1747

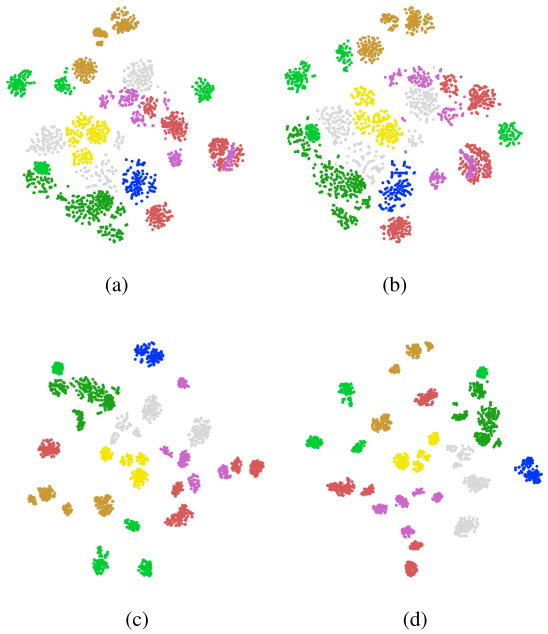


Fig. 13. t-SNE visualization of original and fused HSI-SAR features in Fus-Mamba. (a) Original HSI features. (b) Original SAR features. (c) Fused HSI features after Fus-Mamba. (d) Fused SAR features after Fus-Mamba.

F. Analysis of Fus-Mamba Effects

We aim to understand how the Fus-Mamba block enhances feature representation in multisource remote sensing data by using t-SNE visualizations. The comparison of features before and after fusion provides insights into the strengths of the Fus-Mamba block in integrating diverse data modalities.

Fig. 13 presents the t-SNE visualizations of HSI and SAR features before and after applying the Fus-Mamba block. Fig. 13(a) shows the original HSI features before fusion. Fig. 13(b) presents the original SAR features before fusion. In Fig. 13(a) and (b), the separability between different classes is relatively poor, with noticeable overlap between clusters. This suggests that the original features alone struggle to clearly discriminate between certain categories.

Fig. 13(c) demonstrates the fused HSI features after the Fus-Mamba block is applied. Fig. 13(d) displays the fused SAR features after the fusion process. The visualizations in Fig. 13(c) and (d) show a significant improvement in class distinction, with clearer and more compact clusters. This indicates that the Fus-Mamba block successfully integrates features from different modalities, enhancing the overall representation and improving the discriminative power of the fused features.

G. Analysis of Scanning Route Numbers

The number of scanning routes is an important parameter in our MSFMamba model, as it directly impacts the diversity of spatial feature representations. In this section, we evaluate the effect of different numbers of scanning routes on classification performance and computational efficiency. Specifically, we compared configurations with two, four, and six scanning routes using the Berlin dataset. In the case of two routes, rowwise and columnwise scanning is utilized. In the case of four routes, rowwise, columnwise, reverse rowwise, and reverse columnwise scanning is used. To achieve six routes, two diagonal scanning routes (top left to bottom right and top right to bottom left) are added.

From Table VIII, we observe that the classification performance improves with the addition of reverse scanning directions. The model with four scanning routes achieved the highest performance, indicating that adding more scanning routes enhances the spatial feature diversity and improves the model's ability to differentiate between classes. However, the configuration with six scanning routes, while providing more feature diversity, results in slightly lower performance. This could be due to the increased complexity, which may introduce more noise into the features or lead to overfitting on certain classes. Our analysis shows that using four scanning routes offers the best classification performance in terms of accuracy. These results suggest that adding more scanning routes does not always result in a proportional improvement in performance and may even lead to diminishing returns.

H. Analysis of Computational Complexity

To provide a comprehensive computational complexity analysis, we evaluated the model parameters, floating-point operations per second (FLOPS), and inference time of various methods on the Augsburg dataset, as shown in Table IX. Specifically, we measure the total number of learnable parameters in millions (M), the computational cost in gigaflops (G), and the inference time in seconds (s). Our proposed MSFMamba model exhibits the lowest FLOPs and inference time, confirming its computational efficiency compared to other methods. This advantage highlights MSFMamba's suitability for real-time applications, especially on resource-constrained edge devices such as drones and mobile platforms. Furthermore, MSFMamba requires significantly fewer parameters than most state-of-the-art methods, contributing to reduced memory requirements and facilitating efficient deployment.

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V. Conclusion and Future Work

This article introduced MSFMamba, a novel network designed for the joint classification of multisource remote sensing data. The network utilized SSM-based blocks to achieve linear complexity and a large receptive field. The MSpa-Mamba block efficiently extracted spatial features using a multiscale strategy, reducing computational and feature redundancies. The Spe-Mamba block enhanced spectral feature extraction, enabling effective HSI and LiDAR/SAR fusion, while the Fus-Mamba block facilitated robust cross-modal feature integration by extending the original Mamba design to support dual inputs. Experiments on four datasets demonstrated that MSFMamba achieved state-of-the-art performance, highlighting its robustness, efficiency, and potential for real-world applications.

However, the model has certain limitations, such as reduced accuracy for imbalanced classes and limited interpretability in critical applications such as urban planning and disaster management. Addressing these challenges requires exploring techniques such as dynamic sampling, advanced loss functions, and methods to enhance explainability. Future work will focus on developing hybrid models that combine Mamba with Transformer architectures to further improve long-range feature modeling and multimodal data fusion. In addition, methods for improving the interpretability of MSFMamba will be explored, particularly in terms of how it processes and integrates the contributions of different data modalities. Investigating feature importance and feature selection techniques will also be part of future research, ensuring that the model is both efficient and transparent in its operations.

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